PSGR
Krishnammal College for Women

Affiliated to Bharathiar University \ Autonomous \ College of Excellence \ Accredited with A++ Grade \ Ranked 9th in NIRF 2025

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Peelamedu, Coimbatore – 641004

POLYCYSTIC OVARY SYNDROME UNDIAGNOSED LIKELIHOOD AND RISK FACTOR ANALYSIS

Mini Project Report submitted to PSGR Krishnammal College for women in partial fulfillment of the requirements for the award of the degree of Master of Science in Data Analytics Bharathiar University, Coimbatore – 641046

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ABSTRACT

Polycystic Ovary Syndrome (PCOS) is a common hormonal and metabolic disorder affecting women of reproductive age and is associated with reproductive and metabolic complications. This study focuses on PCOS Prediction and Risk Factor Analysis using a data-driven approach to predict PCOS and identify associated risk factors. The dataset contains clinical, demographic, reproductive, and lifestyle-related attributes such as BMI, menstrual characteristics, insulin resistance, and other health indicators. The data was analyzed using Python for machine learning and Tableau for interactive visualization.

The analysis covers PCOS prediction, model evaluation, risk category analysis, and risk factor analysis. Logistic Regression and Random Forest classification algorithms were applied to predict PCOS diagnosis and evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The predicted PCOS likelihood was further categorized into Low Risk, Medium Risk, and High Risk. Factors such as BMI, menstrual regularity, and insulin resistance were analyzed to identify patterns associated with PCOS risk.

The predicted results and risk categories were integrated into Tableau to develop two dashboards: PCOS Prediction Overview and Risk Factor Analysis. The dashboards use interactive visualizations to present prediction results, risk category distribution, and relationships between important risk factors.

The results demonstrate that machine learning can support PCOS prediction and risk-level analysis. Variations in predicted risk were observed across different clinical, reproductive, and lifestyle-related factors. The analysis provides useful insights into PCOS prediction and associated risk factors.

In conclusion, this study highlights the importance of combining machine learning and data visualization for PCOS prediction and risk assessment. Python enables effective prediction and analysis, while Tableau provides interactive visualization of the results. Overall, the approach supports better understanding of PCOS risk factors and data-driven healthcare decision-making.

1. INTRODUCTION

Polycystic Ovary Syndrome (PCOS) is one of the most common hormonal and metabolic disorders affecting women of reproductive age. It is a complex condition associated with hormonal imbalance, irregular menstrual cycles, changes in ovarian function, and various metabolic and lifestyle-related factors. The symptoms and characteristics associated with PCOS can vary among individuals, making its identification a challenging process. Early identification of individuals who may have a higher likelihood of PCOS can support timely medical evaluation and appropriate healthcare management. Therefore, analyzing health-related data to understand PCOS patterns has become an important application of Data Analytics and Machine Learning.

PCOS-related datasets contain a wide range of demographic, physical, clinical, menstrual, and lifestyle-related attributes. Factors such as age, body mass index, menstrual characteristics, physical activity, dietary habits, and other health indicators can provide valuable information for understanding patterns associated with PCOS. Analyzing these attributes together can help identify relationships between different factors and PCOS diagnosis. However, the presence of multiple variables, missing values, categorical attributes, and imbalanced diagnosis classes makes healthcare data analysis challenging using conventional methods.

Machine Learning provides an effective approach for analyzing complex healthcare datasets and identifying patterns that may not be easily recognized through manual analysis. Classification algorithms can be used to estimate the likelihood of PCOS based on the available attributes. In this project, three Machine Learning algorithms, namely Logistic Regression, Random Forest, and XGBoost, are implemented and compared. The dataset undergoes appropriate preprocessing, including handling missing values and converting categorical variables into numerical form. Since an imbalance may exist between the diagnosis classes, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to improve the representation of the minority class during model training.

Model evaluation is an important stage of the analytical process. The developed classification models are compared using classification performance measures and confusion matrices to understand their predictive behavior. Based on the comparative evaluation, XGBoost is selected as the final model for PCOS likelihood prediction. The selected model is then used to

generate prediction results that can be integrated with the original dataset for further analysis and visualization.

Business Intelligence further enhances the analytical process by presenting the results through interactive visualisations. Tableau is used to develop an interactive dashboard for analyzing PCOS diagnosis patterns and associated risk factors. Visual representations such as charts, KPI cards, filters, and comparative analyses enable users to examine relationships between PCOS diagnosis and factors such as age, BMI, lifestyle characteristics, and other relevant attributes. The interactive nature of the dashboard allows the dataset to be explored from multiple perspectives and provides a clearer understanding of the patterns present in the data.

The system has been developed by integrating Machine Learning and Business Intelligence techniques for healthcare data analysis. It involves loading and analyzing the PCOS dataset, separating input features and the target variable, transforming diagnosis labels into numerical values, splitting the dataset into training and testing subsets, handling missing values, encoding categorical attributes, and applying SMOTE to address class imbalance. Logistic Regression, Random Forest, and XGBoost algorithms are trained and compared to identify the most effective predictive model. Finally, the selected XGBoost model is evaluated to assess its performance in predicting the likelihood of undiagnosed PCOS and identifying key risk factors associated with the condition.

The processed prediction results are exported as a CSV dataset and integrated into Tableau for interactive dashboard development. The dashboard focuses on analyzing PCOS likelihood and risk factor patterns, enabling users to explore the relationship between diagnosis and different demographic, physical, and lifestyle-related attributes. By combining Machine Learning-based likelihood prediction with interactive risk factor analysis, the project demonstrates how healthcare data can be transformed into meaningful analytical information.

The developed solution provides a structured analytical approach for exploring PCOS-related data and identifying patterns within the available dataset. It demonstrates the application of data preprocessing, class balancing, Machine Learning classification, model evaluation, and Business Intelligence visualization in a healthcare analytics context. The project can support exploratory analysis of PCOS-related characteristics and provide a data-driven perspective on the factors associated with PCOS within the dataset.

1.1 LITERATURE REVIEW

Teede et al. [1] (2018) developed international evidence-based guidelines for the assessment and management of Polycystic Ovary Syndrome (PCOS). The study provided standardized recommendations for the diagnosis, treatment, and long-term management of women with PCOS. It emphasized the importance of a comprehensive approach that considers both reproductive and metabolic aspects of the condition.

Skiba et al. [2] (2018) examined variations in the reported prevalence of PCOS across different populations. The study highlighted that differences in diagnostic criteria, study populations, and assessment methods can significantly influence prevalence estimates. This finding demonstrates the challenges involved in consistently identifying PCOS across diverse populations.

Gilbert et al. [3] (2018) reviewed the major comorbidities and complications associated with PCOS. The study emphasized that PCOS is not limited to reproductive abnormalities but is also associated with metabolic and long-term health complications. The findings support the need for early identification and continuous monitoring of individuals at risk.

Pasquali et al. [4] (2018) discussed newer perspectives on the definition and management of PCOS. The study highlighted the heterogeneous nature of the syndrome, where women may exhibit different combinations of reproductive, hormonal, and metabolic characteristics. This variability indicates the importance of considering multiple clinical factors when assessing PCOS.

Carvalho et al. [5] (2018) investigated PCOS as a systemic disorder involving multiple molecular and metabolic pathways. The study described the relationship between PCOS and metabolic abnormalities and emphasized that the condition can affect several physiological systems. These findings support the use of broader approaches for PCOS assessment and risk analysis.

Neven et al. [6] (2018) provided an overview of PCOS diagnostic criteria, prevalence, clinical manifestations, and management strategies. The study emphasized the complexity of diagnosing and managing PCOS due to its diverse clinical presentations. The review also highlighted the importance of appropriate diagnostic assessment for effective management.

1.2 PROBLEM STATEMENT

Polycystic Ovary Syndrome (PCOS) is a complex hormonal and metabolic condition that affects women of reproductive age and is associated with a wide range of demographic, physical, menstrual, and lifestyle-related factors. The variation in symptoms and characteristics among individuals makes it challenging to identify patterns associated with PCOS using individual factors alone. Healthcare datasets related to PCOS contain multiple attributes that may provide valuable information for understanding the characteristics of diagnosed and undiagnosed cases. However, analyzing these attributes manually can be difficult due to the volume, variety, and complexity of the available data.

Traditional approaches such as manual examination, spreadsheets, and static reports may not effectively identify relationships between PCOS diagnosis and factors such as age, BMI, menstrual characteristics, and lifestyle-related attributes. In addition, imbalanced diagnosis classes can affect the performance of Machine Learning classification models by causing the model to favor the majority class. Therefore, appropriate data preprocessing and class-balancing techniques are required before applying predictive models.

Challenges, an integrated Machine Learning and Business Intelligence approach was developed to transform the PCOS dataset into meaningful analytical insights. The proposed "Polycystic Ovary Syndrome Undiagnosed Likelihood Prediction and Risk Factor Analysis" utilizes data preprocessing, SMOTE, and machine learning classification algorithms to predict PCOS likelihood. Logistic Regression, Random Forest, and XGBoost models were trained and compared, and the final XGBoost model was evaluated using classification metrics and a confusion matrix.

The analysis of PCOS-related risk factors is performed through interactive Tableau visualizations. The dashboard enables the exploration of diagnosis patterns across demographic, physical, and lifestyle-related attributes, providing an interactive environment for examining relationships within the dataset. By combining Machine Learning–based likelihood prediction with risk factor analysis, the system provides a structured and data-driven approach for understanding PCOS-related patterns and insights from the available data.

The major problems addressed in the project are:

- Difficulty in analyzing multiple PCOS-related attributes using conventional methods.
- Challenges in identifying patterns and relationships between PCOS diagnosis and associated risk factors.
- Presence of imbalanced diagnosis classes that can affect Machine Learning model performance.
- Need to compare different classification models to identify a suitable model for PCOS likelihood prediction.
- Difficulty in interpreting complex healthcare data through static analysis and reports.
- Need for an interactive dashboard to support effective exploration of PCOS diagnosis patterns and risk factors.

1.3 TOOLS AND TECHNOLOGIES

The "Polycystic Ovary Syndrome Undiagnosed Likelihood Prediction and Risk Factor Analysis" project was developed using a combination of Python-based Data Analytics and Machine Learning technologies along with Tableau for Business Intelligence and interactive data visualization. These tools were used for data preprocessing, Exploratory Data Analysis, class balancing, Machine Learning model development, model evaluation, result generation, and dashboard development. The major tools and technologies used in the project are presented in Table 1.3.1.

Table 1.3.1: Tools and Technologies Used

Tool / Technology	**Purpose**
Python	Data preprocessing, Exploratory Data Analysis, feature engineering, and Machine Learning
Google Colab	Development and execution of Python code
Pandas	Data loading, cleaning, manipulation, and analysis
NumPy	Numerical operations and data processing

Scikit-learn	Data preprocessing, train-test split, model training, and evaluation
Imbalanced-learn (SMOTE)	Handling class imbalance in the training dataset
Logistic Regression	Machine Learning classification model
Random Forest	Machine Learning classification model
XGBoost	Final Machine Learning classification and likelihood prediction
Matplotlib / Seaborn	Exploratory Data Analysis and data visualization
CSV	Storage and transfer of processed Machine Learning results
Tableau	Interactive dashboard development, PCOS risk factor analysis, likelihood analysis, and visualization

Python

Python is a high-level, open-source programming language widely used in Data Analytics and Machine Learning. In this project, Python was used as the primary programming language for loading the PCOS dataset, performing data preprocessing, preparing the input features, implementing Machine Learning algorithms, evaluating model performance, and generating prediction results.

Google Colab

Google Colab are interactive development environments used for writing and executing Python code. They were used to perform dataset understanding, data preprocessing, Exploratory Data Analysis, SMOTE implementation, Machine Learning model training, model comparison, confusion matrix analysis, and final model evaluation.

Pandas and NumPy

Pandas is a Python library used for data manipulation and analysis, while NumPy provides efficient numerical computing capabilities. In this project, Pandas was used to load, clean, manipulate, and analyze the PCOS dataset. NumPy was used to perform numerical operations and support data preprocessing and Machine Learning tasks.

Scikit-learn

Scikit-learn is a Python Machine Learning library that provides tools for data preprocessing, classification, model training, and evaluation. In this project, Scikit-learn was used for train-test splitting, preprocessing operations, implementation of classification models, and evaluation of the developed Machine Learning models.

Imbalanced-learn (SMOTE)

Imbalanced-learn is a Python library that provides techniques for handling imbalanced datasets. The Synthetic Minority Over-sampling Technique (SMOTE) was used to address class imbalance in the training dataset. It generates synthetic samples for the minority class to provide a more balanced dataset for Machine Learning model training.

Machine Learning Algorithms

Three Machine Learning classification algorithms were implemented and compared in the project: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was used as a baseline classification model, while Random Forest was used as an ensemble classification approach. XGBoost was implemented as a powerful gradient boosting classification algorithm and was selected as the final model for PCOS likelihood prediction based on the comparative evaluation.

Matplotlib

Matplotlib is a Python data visualization library used to represent data graphically. In this work, it was utilized for Exploratory Data Analysis (EDA) and data visualization to identify patterns, trends, and relationships within the PCOS dataset.

CSV

Comma-Separated Values (CSV) is a commonly used format for storing and transferring structured tabular data. In this project, the processed Machine Learning results were exported in CSV format and used as the input data for further visualization and dashboard development.

Tableau

Tableau is a Business Intelligence and data visualization platform used to transform data into interactive dashboards. In this project, Tableau was used to develop an interactive dashboard for PCOS likelihood analysis and risk factor analysis. The dashboard enables users to explore PCOS diagnosis patterns across different demographic, physical, and lifestyle-related attributes using KPI cards, charts, filters, and other interactive visualizations.

The integration of Python, Pandas, NumPy, Scikit-learn, Imbalanced-learn, Machine Learning algorithms, Matplotlib, Seaborn, CSV, and Tableau provides a complete analytical workflow for the project. Python-based tools were primarily used for data preparation, Exploratory Data Analysis, class balancing, Machine Learning model development, and evaluation, while Tableau was used to present the processed results through interactive dashboards. This integrated approach transforms the raw PCOS dataset into meaningful predictive and visual analytical information.

PCOS Dataset

The dataset used in this project was obtained from the Kaggle platform and contains approximately 2,40,000 instances with 17 attributes related to Polycystic Ovary Syndrome (PCOS). The dataset includes information related to demographic characteristics, physical attributes, clinical and menstrual factors, lifestyle-related factors, and PCOS diagnosis. The dataset provides a suitable foundation for analyzing PCOS-related patterns, identifying associated risk factors, and developing a Machine Learning-based likelihood prediction model.

The integration of Python, Pandas, NumPy, Scikit-learn, Imbalanced-learn, Google Colab, and Tableau combines Machine Learning with Business Intelligence to transform raw PCOS-related data into meaningful healthcare insights. Each technology performs a specific role, beginning with data understanding and preprocessing in Python, followed by class balancing using SMOTE, classification using Logistic Regression, Random Forest, and XGBoost, and evaluation of the developed models. The processed Machine Learning results are exported as a CSV file and integrated into Tableau for interactive visualization and dashboard development.

This integrated workflow enables the analysis of PCOS diagnosis patterns, likelihood of PCOS, and associated risk factors, providing valuable insights into the relationships between

PCOS diagnosis and relevant demographic, physical, clinical, menstrual, and lifestyle-related attributes. The developed solution supports exploratory healthcare analysis and presents the findings through interactive and easy-to-understand visualizations.

1.4 DOMAIN – HEALTHCARE ANALYTICS

Healthcare Analytics is a specialized domain that focuses on collecting, processing, analyzing, and interpreting healthcare-related data to identify meaningful patterns and support data-driven decision-making. The increasing availability of healthcare datasets has created opportunities to apply Data Analytics, Machine Learning, and Business Intelligence techniques to understand health conditions, identify associated factors, and develop analytical solutions for healthcare-related problems.

Healthcare Analytics is applied to the analysis of Polycystic Ovary Syndrome (PCOS) using data-driven techniques. PCOS is a complex hormonal and metabolic condition influenced by various demographic, physical, clinical, menstrual, and lifestyle-related factors. Analyzing these factors helps identify patterns, trends, and relationships associated with PCOS diagnosis within the dataset. The integration of multiple attributes enables the application of analytical methods to gain deeper insights into PCOS-related characteristics and support data-driven understanding of the condition.

Machine Learning is an important component of Healthcare Analytics that enables the identification of patterns within complex healthcare datasets. Classification algorithms can be applied to estimate the likelihood of a particular health condition based on available input attributes. In this project, Logistic Regression, Random Forest, and XGBoost are implemented and compared for PCOS likelihood prediction. The Synthetic Minority Over-sampling Technique (SMOTE) is applied to address class imbalance in the training dataset and improve the representation of the minority class during model training.

Business Intelligence further enhances healthcare data analysis by transforming processed data and Machine Learning results into interactive visualizations. Tableau is used in this project to develop an interactive dashboard for analyzing PCOS diagnosis patterns and associated risk factors. The dashboard enables users to explore relationships between PCOS diagnosis and relevant attributes through KPI cards, charts, filters, and other interactive visual components.

Applications of Healthcare Analytics

- Analysis of health-related datasets to identify meaningful patterns.
- Identification and analysis of factors associated with specific health conditions.
- Machine Learning-based classification and likelihood prediction.
- Analysis of demographic, physical, clinical, and lifestyle-related characteristics.
- Interactive visualization of healthcare data using Business Intelligence tools.
- Supporting data-driven exploratory analysis and healthcare decision-making.

Healthcare Analytics provides an effective framework for transforming complex healthcare datasets into meaningful analytical information. In this project, the integration of Data Analytics, Machine Learning, and Business Intelligence enables PCOS likelihood prediction and risk factor analysis. The developed analytical approach demonstrates how healthcare data can be processed, analyzed, modelled, and visualized to provide clear and interactive insights into PCOS-related patterns.

2.DATA MODELLING

2.1 PROCESS FLOW

The development of the “Polycystic Ovary Syndrome Undiagnosed Likelihood and Risk Factor Analysis” follows a systematic workflow that integrates data preparation, Exploratory Data Analysis, Machine Learning, and Business Intelligence techniques. Each stage is carefully designed to ensure the quality, consistency, and reliability of the analytical process, enabling the transformation of the raw PCOS dataset into meaningful prediction results and healthcare insights. The structured workflow establishes a logical sequence of activities in which each phase contributes to the development of the final interactive Tableau dashboard.

The process begins with Data Collection, where the PCOS dataset is obtained and loaded into the Python environment. The dataset is then examined during the Data Understanding and Validation stage to identify its structure, attributes, data types, missing values, duplicate records, and target-class distribution. This stage ensures that the dataset is suitable for further analytical processing.

During the Data Pre-processing stage, the input and target variables are separated, and the diagnosis values are converted from Yes/No into binary numerical values (1/0). Missing values are handled, categorical variables are converted into numerical representations, and the dataset is divided into training and testing sets. These preprocessing operations prepare the dataset for reliable Machine Learning model development.

The prepared dataset is subsequently examined through Exploratory Data Analysis (EDA) to understand the distribution and characteristics of the variables. Numerical summaries, target distribution, categorical analysis, correlation analysis, and visualizations are used to identify patterns and relationships within the dataset. Feature Engineering is then performed to prepare the relevant attributes in a suitable format for model development.

Since the target classes may not be equally represented, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training data to address class imbalance. This improves the representation of the minority class and provides a balanced dataset for Machine Learning model training.

The balanced dataset is then used for Machine Learning Model Development, where three classification algorithms, namely Logistic Regression, Random Forest, and XGBoost, are trained and evaluated. The models are compared using evaluation measures such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Detailed evaluation and threshold analysis are performed to assess the classification performance and determine the final model used for generating PCOS likelihood predictions.

The final prediction results are used for Risk/Likelihood Analysis, where the predicted PCOS likelihood is interpreted and categorized into different risk levels. The prediction results are combined with the relevant patient attributes to create a structured Machine Learning result dataset. This dataset is exported in CSV format and integrated into Tableau for interactive dashboard development.

The Tableau dashboards provide visual analysis of PCOS likelihood, diagnosis distribution, risk categories, demographic characteristics, lifestyle-related factors, BMI-related patterns, and other risk factors. Interactive filters, charts, KPI elements, and analytical visualizations enable users to explore the dataset and identify meaningful patterns associated with PCOS.

The process flow provides a structured approach for implementing the project by ensuring that each stage is completed in a logical sequence. Data preparation and exploratory analysis establish a reliable foundation for Machine Learning, while the classification models generate predictions that can be further interpreted through likelihood and risk analysis. The resulting dataset is then visualized through Tableau, enabling users to analyze PCOS-related patterns and risk factors efficiently. This systematic workflow improves the consistency, interpretability, and effectiveness of the overall PCOS analytics solution.

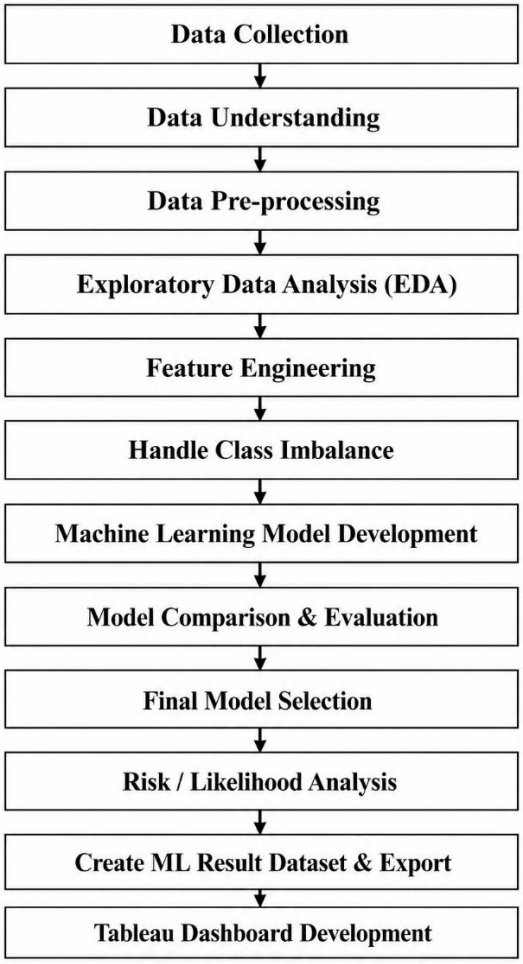


Figure 2.1: Process Flow

Figure 2.1 illustrates the complete workflow adopted in the project. The diagram demonstrates the logical progression from data collection and validation through data preprocessing, EDA, feature engineering, class-imbalance handling, Machine Learning, model evaluation, risk/likelihood analysis, and CSV export, followed by interactive Tableau dashboard development.

2.2 DATA COLLECTION

The dataset used for the “Polycystic Ovary Syndrome Undiagnosed Likelihood Prediction and Risk Factor Analysis” project was obtained from the Kaggle platform. It consists of approximately 2,40,000 instances and 17 attributes related to Polycystic Ovary Syndrome (PCOS). The dataset contains a combination of demographic, physical, clinical, menstrual, lifestyle, and

diagnosis-related attributes that provide a suitable foundation for analyzing PCOS-related patterns and developing a Machine Learning-based likelihood prediction model.

The collected dataset serves as the primary source for analyzing PCOS diagnosis patterns, estimating the likelihood of PCOS, and examining associated risk factors. After collection, the dataset was imported into Google Colab using Python for data understanding and preprocessing. The dataset was examined, and the input features and target variable were separated. The diagnosis values were converted from Yes/No into 1/0, followed by train-test splitting, handling of missing values, and conversion of categorical attributes into numerical form. SMOTE was then applied to the training dataset to address class imbalance.

The prepared dataset was used to train and compare three Machine Learning classification models: Logistic Regression, Random Forest, and XGBoost. The models were evaluated using classification performance measures and confusion matrix analysis. Based on the comparative evaluation, XGBoost was selected as the final model for PCOS likelihood prediction. The processed prediction results were subsequently exported as a CSV file and integrated into Tableau for interactive dashboard development and PCOS risk factor analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Country	Age	BMI	Menstrual Regularity	Hirsutism	Acne Severity	Family History	Insulin Resistance	Lifestyle Score	Stress Levels	Urban/Rural	Socioeconomic	Awareness	Fertility Concern	Undiagnosed PCOS	Ethnicity	Diagnosis
2	Madagascar	26	Overweight	Regular	Yes	Severe	Yes	Yes	2	Low	Rural	High	Yes	No	0.107938174	Hispanic	Yes
3	Vietnam	16	Underweight	Regular	Yes	None	No	Yes	4	High	Rural	Middle	Yes	No	0.156729271	Other	No
4	Somalia	41	Normal	Regular	No	Moderate	No	No	7	Medium	Urban	Middle	Yes	Yes	0.202900736	Other	No
5	Malawi	27	Normal	Irregular	No	Mild	No	No	10	Low	Urban	High	Yes	No	0.073925956	Caucasian	Yes
6	France	26	Overweight	Irregular	Yes	None	No	No	7	Medium	Urban	Middle	No	No	0.229266156	Caucasian	No
7	Rwanda	17	Overweight	Irregular	No	None	No	Yes	3	Low	Urban	High	Yes	Yes	0.209841236	African	No
8	Rwanda	39	Obese	Regular	Yes	Mild	No	Yes	10	Medium	Urban	Low	No	Yes	0.208061775	African	No
9	Tanzania	38	Normal	Regular	Yes	Mild	No	No	8	Medium	Rural	Middle	No	No	0.205956516	Caucasian	No
10	United States	47	Normal	Regular	Yes	None	Yes	No	1	Medium	Urban	Low	Yes	No	0.103628129	Asian	Yes
11	Italy	21	Underweight	Irregular	No	Mild	No	No	1	Medium	Rural	Low	Yes	No	0.102774384	African	No
12	Australia	25	Normal	Irregular	Yes	Moderate	Yes	No	5	Low	Rural	Low	Yes	Yes	0.181859768	Asian	No
13	India	20	Normal	Regular	No	None	Yes	No	2	Medium	Urban	Middle	Yes	Yes	0.180212805	Caucasian	No
14	Argentina	46	Underweight	Regular	No	Mild	No	Yes	1	Medium	Urban	Low	No	Yes	0.232999434	Caucasian	No
15	Morocco	39	Normal	Irregular	Yes	None	No	No	3	Medium	Urban	High	Yes	No	0.167554043	Hispanic	No
16	India	19	Overweight	Regular	Yes	Mild	No	No	9	Medium	Urban	Middle	Yes	No	0.116934506	Caucasian	No
17	Zambia	26	Normal	Regular	Yes	None	Yes	Yes	4	High	Urban	High	Yes	No	0.114724588	African	No
18	Romania	35	Normal	Regular	No	None	No	No	3	High	Urban	High	Yes	Yes	0.126046274	Caucasian	No
19	France	48	Overweight	Regular	Yes	None	No	No	8	Low	Urban	Middle	No	No	0.224517257	African	No
20	Sudan	22	Overweight	Regular	Yes	Mild	No	No	6	Medium	Urban	Low	Yes	No	0.061689463	Hispanic	No
21	Benin	29	Underweight	Regular	No	Moderate	No	No	3	Medium	Rural	Low	Yes	No	0.228870342	Hispanic	No
22	Burkina Faso	31	Overweight	Regular	No	Mild	Yes	No	1	High	Rural	Middle	Yes	No	0.051418218	Caucasian	No
23	Italy	20	Overweight	Regular	No	Mild	No	No	10	High	Urban	High	Yes	Yes	0.197915819	Asian	No
24	Nepal	36	Normal	Irregular	Yes	None	No	No	3	Low	Urban	Low	Yes	No	0.155153512	Hispanic	No
25	Mali	37	Normal	Regular	Yes	None	No	No	3	Medium	Urban	Low	Yes	No	0.105761601	Asian	No
26	Malaysia	20	Obese	Regular	Yes	Mild	No	No	9	Low	Urban	Middle	Yes	Yes	0.239350082	Asian	No
27	Romania	34	Normal	Regular	Yes	None	No	Yes	8	Medium	Urban	High	Yes	No	0.145244483	Other	No

Figure 2.2: Sample of the PCOS Dataset

Figure 2.2 illustrates a sample of the PCOS dataset used in this project. It illustrates the structured format of the dataset and the different attributes used for analysis and Machine Learning. The dataset provides the required information for data preprocessing, model development, PCOS likelihood prediction, and interactive risk factor analysis through Tableau.

2.3 DATASET DESCRIPTION

The PCOS Prediction Dataset consists of 240,000 records and 17 attributes representing demographic, physical, reproductive, lifestyle, socioeconomic, and health-related characteristics associated with Polycystic Ovary Syndrome (PCOS). The dataset contains both numerical and categorical information that provides the required foundation for analyzing PCOS-related patterns, identifying potential risk factors, and predicting the likelihood of undiagnosed PCOS.

The dataset includes attributes such as Country, Age, BMI, Menstrual Regularity, Hirsutism, Acne Severity, Family History, Insulin Resistance, Lifestyle Score, Stress Levels, Urban/Rural, Socioeconomic, Awareness, Fertility Concerns, Undiagnosed PCOS, Ethnicity, and Diagnosis. These attributes enable the analysis of demographic, physical, reproductive, lifestyle, and health-related factors associated with PCOS.

The dataset was used for data validation, data pre-processing, Exploratory Data Analysis, feature engineering, Machine Learning model development, likelihood prediction, and risk factor analysis. The resulting prediction data was exported in CSV format and integrated into Tableau for interactive dashboard development and visualization.

Table 2.3: Dataset Information

Attribute	**Description**
Dataset Name	PCOS Prediction Dataset
Records	240,000
Attributes	17
Domain	Healthcare / PCOS Risk Factor Analysis
File Format	CSV
Target Variable	Diagnosis
Prediction Variable	Undiagnosed PCOS

Table 2.3 illustrates the general characteristics of the PCOS Prediction Dataset, including the dataset name, total number of records, number of attributes, application domain, file format, target variable, and prediction variable.

Table 2.4: Classification of Dataset Attributes

Category	**Attributes**	**Purpose**
Demographic Details	Country, Age, Ethnicity	Represents the demographic characteristics of individuals.
Physical Characteristics	BMI	Represents the individual's BMI category.
Menstrual and Reproductive Factors	Menstrual Regularity, Fertility Concerns	Represents menstrual patterns and reproductive-related characteristics.
PCOS Symptoms	Hirsutism, Acne Severity	Represents physical symptoms associated with PCOS.
Medical and Family Factors	Family History, Insulin Resistance	Represents hereditary and metabolic factors associated with PCOS.
Lifestyle and Stress Factors	Lifestyle Score, Stress Levels	Represents lifestyle characteristics and stress levels.
Socioeconomic and Environmental Factors	Urban/Rural, Socioeconomic	Represents socioeconomic and residential characteristics.
Awareness	Awareness	Represents awareness related to PCOS.
Prediction	Undiagnosed PCOS	Represents the predicted likelihood of undiagnosed PCOS.
Target	Diagnosis	Represents the actual PCOS diagnosis status (Yes/No).

Table 2.4 illustrates the 17 dataset attributes into functional categories such as demographic details, physical characteristics, reproductive factors, PCOS symptoms, medical and family factors, lifestyle and stress factors, socioeconomic characteristics, prediction information, and the target variable. This classification provides a structured understanding of the dataset and supports systematic risk factor analysis.

The categorization of dataset attributes provides a clear understanding of the information available for analysis. The combination of numerical and categorical attributes enables comprehensive examination of demographic, physical, reproductive, lifestyle, and health-related factors. The Undiagnosed PCOS variable provides likelihood information, while the Diagnosis variable serves as the target for Machine Learning classification. The organized dataset structure supports subsequent Machine Learning analysis and interactive Tableau dashboard development.

2.4 DATA PRE-PROCESSING

Data pre-processing is an essential stage that prepares the PCOS dataset for Machine Learning and risk factor analysis. During this stage, the dataset was examined to verify its structure, data types, statistical characteristics, missing values, and duplicate records. The required preprocessing techniques were then applied to convert the raw dataset into a suitable format for Machine Learning model development.

Dataset Structure

The dataset structure was examined to understand the number of records, attributes, and overall organization of the dataset before further analysis.

```
Dataset loaded successfully!
Shape: (240000, 17)
```

```
df.head()
```

	Country	Age	BMI	Menstrual Regularity	Hirsutism	Acne Severity	Family History of PCOS	Insulin Resistance	Lifestyle Score	Stress Levels	Urban/Rural	Socioeconomic Status	Awareness of PCOS	Fertility Concerns	Undiagnosed PCOS Likelihood	Ethnici
0	Madagascar	26.0	Overweight	Regular	Yes	Severe	Yes	Yes	2.0	Low	Rural	High	Yes	No	0.107938	Hispa
1	Vietnam	16.0	Underweight	Regular	Yes	NaN	No	Yes	4.0	High	Rural	Middle	Yes	No	0.156729	Oti
2	Somalia	41.0	Normal	Regular	No	Moderate	No	No	7.0	Medium	Urban	Middle	Yes	Yes	0.202901	Oti
3	Malawi	27.0	Normal	Irregular	No	Mild	No	No	10.0	Low	Urban	High	Yes	No	0.073926	Caucas
4	France	26.0	Overweight	Irregular	Yes	NaN	No	No	7.0	Medium	Urban	Middle	No	No	0.229266	Caucas

```
print("Number of rows:", df.shape[0])
```

Figure 2.5: Dataset Structure

Figure 2.5 illustrates the structure of the PCOS dataset containing 240,000 records and 17 attributes. The attributes provide information related to demographic, physical, reproductive, lifestyle, socioeconomic, and health-related characteristics used for PCOS analysis.

Data Type Verification

The data types of all attributes were verified to distinguish numerical variables from categorical variables and to determine the appropriate preprocessing techniques.

```
•••  Country                        object
     Age                           float64
     BMI                            object
     Menstrual Regularity           object
     Hirsutism                      object
     Acne Severity                  object
     Family History of PCOS         object
     Insulin Resistance             object
     Lifestyle Score               float64
     Stress Levels                  object
     Urban/Rural                    object
     Socioeconomic Status           object
     Awareness of PCOS              object
     Fertility Concerns             object
     Undiagnosed PCOS Likelihood   float64
     Ethnicity                      object
     Diagnosis                      object
     dtype: object
```

Figure 2.6: Data Types of Dataset Attributes

Figure 2.6 illustrates the data types of the attributes available in the PCOS dataset. Numerical and categorical attributes are identified to ensure appropriate processing before Machine Learning.

Statistical Summary

Descriptive statistics were examined to understand the numerical characteristics and distribution of the dataset.

```
Diagnosis distribution:
Diagnosis
No     147411
Yes     92589
Name: count, dtype: int64

Diagnosis percentage:
Diagnosis
No     61.42125
Yes    38.57875
Name: proportion, dtype: float64
```

Figure 2.7: Statistical Summary of the Dataset

Figure 2.7 illustrates statistical measures such as count, mean, standard deviation, minimum, maximum, and quartile values for the numerical attributes. This provides an initial understanding of the numerical characteristics of the dataset.

Missing Value Analysis

Dataset was checked for missing values to ensure data completeness and identify attributes requiring further treatment.

```
Missing values:
Acne Severity    60085
dtype: int64
```

Figure 2.8: Missing Value Analysis

Figure 2.8 illustrates the results of the missing value analysis performed on the PCOS dataset. The identified missing values were handled during the preprocessing stage using appropriate imputation techniques.

Duplicate Value Analysis

Dataset was examined for duplicate records to ensure that repeated observations did not affect the subsequent analysis.

```
Duplicate rows: 0
```

Figure 2.9: Duplicate Value Analysis

Figure 2.9 illustrates the duplicate record analysis performed on the dataset. The analysis was used to verify the presence of repeated observations before further processing.

Target Variable Preparation

The diagnosis attribute was selected as the target variable for the classification models. The Yes and No values were converted into binary numerical values to make the target variable suitable for Machine Learning.

```
Diagnosis
0    147411
1     92589
Name: count, dtype: int64

Missing target values: 0
```

Figure 2.10: Target Variable Preparation

Figure 2.10 illustrates the preparation of the diagnosis target variable. The distribution of the diagnosis classes was examined before applying the class imbalance handling technique.

Train-Test Split

Dataset was divided into training and testing subsets before Machine Learning model development. A 70:30 ratio was used for training and testing, respectively.

```
Training data: (168000, 16)
Testing data: (72000, 16)
```

Figure 2.11: Train-Test Split

Figure 2.11 illustrates the division of the dataset into training and testing sets. The training dataset was used for model development, while the testing dataset was retained for model evaluation.

Categorical Variable Encoding

Categorical attributes were converted into numerical representations using one-hot encoding so that they could be processed by the selected Machine Learning algorithms.

```
X_train_encoded = X_train_encoded.astype(float)
X_test_encoded = X_test_encoded.astype(float)

print(
    X_train_encoded.dtypes.value_counts()
)
```

```
float64    99
Name: count, dtype: int64
```

Figure 2.12: Categorical Variable Encoding

Figure 2.12 illustrates the transformation of categorical attributes into numerical features. The encoded training and testing datasets were aligned to maintain a consistent feature structure.

Final Data Verification

The processed datasets were finally checked for remaining missing and infinite values to ensure data quality before Machine Learning model development.

```
NaN in X_train: 0
NaN in X_test: 0
Infinity in X_train: 0
Infinity in X_test: 0
```

Figure 2.13: Final Data Verification

Figure 2.13 illustrates the final verification of the pre-processed dataset. The verification ensures that the processed training and testing datasets are complete and suitable for the subsequent Machine Learning stage.

The data pre-processing stage established a structured and consistent foundation for the subsequent Machine Learning analysis. The processed dataset was then used for class imbalance handling using SMOTE and development of the selected classification models.

Handling Class Imbalance

Class imbalance was examined in the diagnosis target variable before developing the Machine Learning models. An imbalanced target distribution may cause a classification model to favor the majority class and reduce its ability to correctly identify the minority class. Therefore, a suitable oversampling technique was applied to the training dataset to obtain a balanced class distribution.

Class Distribution Analysis

The distribution of the diagnosis classes was examined before applying the class balancing technique to identify the difference between the majority and minority classes.

```
Before SMOTE:
Diagnosis
0    103188
1     64812
Name: count, dtype: int64

Percentage:
Diagnosis
0    61.421429
1    38.578571
Name: proportion, dtype: float64
```

Figure 2.14: Class Distribution Before SMOTE

Figure 2.14 illustrates the distribution of the diagnosis classes before applying SMOTE. The results show the number of observations belonging to each class and indicate the presence of class imbalance in the training dataset.

SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training dataset to address the identified class imbalance. The technique generates synthetic samples for the minority class, thereby improving the balance between the diagnosis classes without directly duplicating existing observations.

```
Before SMOTE:
Diagnosis
0    103188
1     64812
Name: count, dtype: int64

After SMOTE:
Diagnosis
0    103188
1    103188
Name: count, dtype: int64
```

Figure 2.15: Class Distribution After SMOTE

Figure 2.15 illustrates the class distribution after applying SMOTE to the training data. The results show a balanced representation of the diagnosis classes, providing a suitable training dataset for developing the Logistic Regression, Random Forest, and XGBoost classification models.

3.PCOS PREDICTION ANALYSIS

3.1 XGBOOST – MODEL CREATION

XGBoost was implemented as a gradient-boosting classification algorithm for PCOS prediction. The model builds a sequence of decision trees in which subsequent trees improve upon the errors of previous trees. In this project, XGBoost was trained using the SMOTE-balanced training dataset and used to generate both predicted classes and probability scores.

The probability scores generated by XGBoost were particularly important for the subsequent stages of the project, as they were used for threshold optimisation and the classification of observations into different PCOS risk categories. The XGBoost model was further optimized through hyperparameter tuning to obtain an improved model for the final prediction process.

Formulae:

$$\hat{y}_i = \sum_{k=1}^{K} f_k\,(X_i)$$

Table 3. 1: Description of XGBoost Prediction Formula

Symbol	Description
$\hat{y}_i$	Predicted output for observation i
K	Total number of decision trees
f_k	The k^{th}decision tree
X_i	Input features for observation i

Table 3.1 illustrates the symbols used in the XGBoost prediction formulation. The model combines the outputs of multiple decision trees to produce the final prediction. Each tree contributes to the overall prediction, allowing XGBoost to progressively improve the classification result.

XGBOOST MODEL OPTIMIZATION

After developing the initial classification models, the XGBoost model was further optimized to identify a suitable combination of hyperparameters for the PCOS prediction task. Hyperparameter optimization was performed to improve the model configuration and obtain a suitable model for generating the final prediction probabilities. The optimization process was carried out using the SMOTE-balanced training dataset.

Randomized search

Randomized Search was used to identify suitable hyperparameter combinations for the XGBoost classifier. The parameter search included n_estimators, max_depth, learning_rate, subsample, colsample_bytree, min_child_weight, and gamma. These parameters control different aspects of the model, such as the number of trees, tree depth, learning rate, data sampling, and regularisation.

The RandomizedSearchCV method was configured to evaluate 15 randomly selected parameter combinations using three-fold cross-validation. The F1-score was used as the scoring criterion for selecting the suitable parameter configuration. The search was performed using the SMOTE-balanced training dataset.

The Randomized Search process resulted in a best estimator and corresponding set of hyperparameters, which were used for developing the tuned XGBoost model.

Best Xgboost Parameters

The best estimator obtained from the Randomized Search process was selected using the best_estimator_ attribute. The corresponding hyperparameters were obtained using the best_params_ attribute. These parameters represent the configuration selected by the optimization process based on the specified F1-score evaluation criterion.

```
Best XGBoost Parameters:
{'subsample': 0.9, 'n_estimators': 300, 'min_child_weight': 1, 'max_depth': 6, 'learning_rate': 0.1, 'gamma': 0.1, 'colsample_bytree': 0.9}
```

Figure 3.2: Best XGBoost Parameters Obtained Through Randomized Search

Figure 3.2 illustrates the hyperparameter configuration selected through the Randomized Search process. The identified parameters were used to obtain the tuned XGBoost estimator for the subsequent prediction and evaluation stages.

Tuned Xgboost Prediction

The best XGBoost estimator obtained from the optimization process was used to generate predictions for the encoded test dataset. Two types of outputs were generated: predicted class labels and prediction probabilities. The predicted class labels were stored in xgb_pred, while the probability corresponding to the positive Diagnosis class was stored in xgb_prob.

The prediction probabilities generated by the tuned XGBoost model were retained for the subsequent threshold optimization process. These probability values were also used later for generating the PCOS risk categories.

Threshold Optimization

The prediction probabilities generated by the tuned XGBoost model were further analyzed to identify a suitable classification threshold for the PCOS prediction task. Threshold optimization was performed by evaluating multiple threshold values and measuring the resulting classification performance. This approach helped determine a threshold that provided a suitable balance between precision and recall.

Threshold Evaluation

Threshold values ranging from 0.10 to 0.90 were evaluated at intervals of 0.05. For each threshold, the XGBoost prediction probabilities were converted into binary class predictions. Accuracy, precision, recall, and F1-score were calculated for every evaluated threshold.

The F1-score was used as the primary criterion for selecting the best threshold because it provides a balance between precision and recall.

The threshold-based classification can be represented as:

$$\hat{y} = \begin{cases} 1, & \text{if } P \geq t \\ 0, & \text{if } P < t \end{cases}$$

where Prepresents the predicted probability and $t$represents the selected threshold.

	Threshold	Accuracy	Precision	Recall	F1 Score
0	0.10	0.729889	0.597609	0.917918	0.723915
1	0.15	0.904000	0.883139	0.865716	0.874341
2	0.20	0.925778	0.944130	0.858408	0.899231
3	0.25	0.930833	0.961198	0.855240	0.905128
4	0.30	0.932389	0.968276	0.852684	0.906811
5	0.35	0.933375	0.972723	0.851172	0.907897
6	0.40	0.934000	0.976294	0.849552	0.908524

Figure 3.3: Classification Performance at Different Threshold Values

Figure 3.3 illustrates the classification performance obtained at different probability thresholds. The table includes the threshold value along with the corresponding accuracy, precision, recall, and F1-score. The results were compared to identify the threshold that provided the highest F1-score.

Best Threshold Selection

The best threshold was identified by locating the threshold corresponding to the maximum F1-score in the threshold evaluation results. The threshold with the highest F1-score was selected as the optimal threshold for generating the final class predictions.

The selected threshold was stored as optimal_threshold and subsequently applied to the XGBoost prediction probabilities.

```
Best Threshold:
Threshold    0.500000
Accuracy     0.935264
Precision    0.983234
Recall       0.846636
F1 Score     0.909837
Name: 8, dtype: float64
```

Figure 3.4: Best Threshold

Figure 3.4 illustrates the threshold configuration that achieved the highest F1-score among the evaluated threshold values. The selected threshold was used to convert the XGBoost probability outputs into the final optimized class predictions.

Optimized Threshold Prediction

The selected optimal threshold was applied to the XGBoost prediction probabilities to generate the final prediction labels. When the predicted probability was greater than or equal to the selected threshold, the observation was classified as the positive class; otherwise, it was classified as the negative class.

The optimized predictions were stored in final_prediction and were subsequently used for calculating the final classification performance.

```
Optimal Threshold: 0.5
```

Figure 3.5: Optimal Threshold and Final Prediction

Figure 3.5 illustartes the optimal threshold used to generate the final XGBoost predictions. The selected threshold was applied to the predicted probabilities, producing the optimized classification results used for the final model evaluation.

Final Xgboost Performance

The final performance of the optimized XGBoost model was evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. These metrics provide a comprehensive assessment of the model's classification capability after applying the optimized threshold.

```
FINAL XGBOOST PERFORMANCE
-------------------------
Accuracy : 0.9353
Precision: 0.9832
Recall   : 0.8466
F1 Score : 0.9098
ROC-AUC  : 0.9335
```

Figure 3.6: Final XGBoost Performance

Figure 3.6 illustrates the final performance measures obtained from the optimized XGBoost model. Accuracy represents the overall proportion of correctly classified observations, while precision measures the correctness of positive predictions. Recall indicates the model's ability to identify positive observations, and F1-score provides a balance between precision and recall. ROC-AUC represents the model's ability to distinguish between the two Diagnosis classes based on the predicted probabilities.

3.2 RISK CATEGORY ANALYSIS

The XGBoost classification model generates a predicted probability of PCOS for each observation. These probability values provide an estimate of the likelihood of PCOS rather than simply assigning a Yes/No prediction. To make these probability values easier to interpret, they are grouped into three risk categories: Low Risk, Medium Risk, and High Risk.

The risk categories are created using predefined probability thresholds. This converts the continuous predicted probability into a categorical variable that can be easily interpreted and used for further analysis.

Risk Category Classification

The predicted probabilities generated by the XGBoost model are stored in xgb_prob. Since the probability values range from 0 to 1, three probability intervals are defined using the pd.cut() function.

- 0.00–0.30 → Low Risk
- 0.30–0.70 → Medium Risk
- 0.70–1.00 → High Risk

The bins parameter specifies the boundaries of the three intervals, while the labels parameter assigns the corresponding risk category. The include_lowest=True parameter ensures that the lowest probability value is included in the first interval.

Risk Category Assignment

The pd.cut() function examines each predicted probability and assigns it to the appropriate interval. For example, a predicted probability below or equal to the low-risk threshold is assigned

Low Risk, while probabilities within the intermediate range are assigned Medium Risk. Probabilities falling in the highest range are assigned High Risk.

This process transforms the numerical probability generated by the XGBoost model into an understandable risk-level classification. The resulting risk_category variable represents the predicted risk level for each observation.

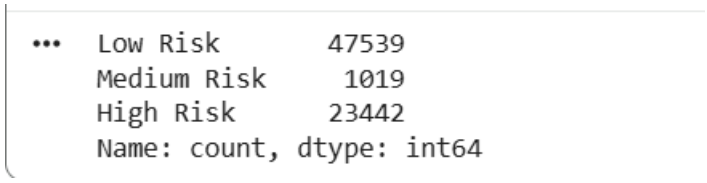


Figure 3.7: Distribution of Predicted PCOS Risk Categories

Figure 3.7 illustrates the distribution of individuals across the three predicted PCOS risk categories generated from the XGBoost model probabilities. The results show 47,539 individuals in the Low Risk category, 1,019 individuals in the Medium Risk category, and 23,442 individuals in the High Risk category. Among the three categories, Low Risk contains the highest number of individuals, while Medium Risk contains the lowest. This distribution shows how the predicted probability values are grouped into distinct risk levels based on the predefined probability thresholds.

Risk Category Distribution

The frequency of each risk category is obtained using the value_counts() function. The resulting output is:

Table 3.8: Risk Category Distribution

Risk Category	Number of Observations
Low Risk	47,539
Medium Risk	1,019
High Risk	23,442
Total	72,000

Table 3.8 illustrates the distribution of observations across the three predicted PCOS risk categories based on the XGBoost probability scores. The Low Risk category has the highest number of observations (47,539), followed by High Risk (23,442), while Medium Risk contains 1,019 observations.

The results show that out of 72,000 observations, 47,539 observations are classified as Low Risk, 1,019 observations are classified as Medium Risk, and 23,442 observations are classified as High Risk.

The Low Risk category has the highest number of observations, followed by the High Risk category. The Medium Risk category contains the smallest number of observations.

Interpretation

The distribution indicates that the predicted probabilities are concentrated mainly in the lower and higher probability ranges. A large number of observations have predicted probabilities within the Low Risk range, while a substantial number fall within the High Risk range.

The relatively small number of Medium Risk observations indicates that fewer observations have predicted probabilities between 30% and 70%. Therefore, the selected thresholds produce three clearly differentiated risk groups, with the majority of observations belonging to either the Low Risk or High Risk category.

The risk categories provide a simplified interpretation of the probability generated by the XGBoost model. Instead of interpreting an individual probability value for every observation, the observations can be viewed according to their corresponding risk level.

3.3 MODEL EVALUATION

The developed Machine Learning models were evaluated to determine their effectiveness in predicting the Diagnosis outcome. Logistic Regression, Random Forest, and XGBoost were considered during the modelling process. The evaluation was performed using classification metrics and a confusion matrix to obtain a detailed understanding of the prediction performance.

LOGISTIC REGRESSION

Logistic Regression is a supervised machine learning classification algorithm used to predict the probability of an observation belonging to a particular class. In this study, Logistic Regression was used to predict the Diagnosis outcome, where the observations are classified into the respective Diagnosis classes. The model establishes a relationship between the input features and the target variable and produces a probability value for classification.

The probability of the positive class is calculated using the Sigmoid function:

$$\mathbf{P(Y = 1) = \frac{1}{1 + e^{-z}}}$$

$$\mathbf{z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n}$$

Here, $P(Y = 1)$represents the probability of the positive Diagnosis class, $\beta_0$represents the intercept, $\beta_1, \beta_2, \ldots, \beta_n$are the model coefficients, and $X_1, X_2, \ldots, X_n$represent the input features.

The sigmoid function converts the model output into a probability ranging from 0 to 1, which is then used to determine the predicted Diagnosis class. Logistic Regression was considered as one of the classification models for comparison with Random Forest and Tuned XGBoost.

RANDOM FOREST

Random Forest is a supervised machine learning ensemble classification algorithm that combines the predictions of multiple decision trees to produce a final classification result. In this study, Random Forest was used to predict the Diagnosis outcome based on the available input features. The use of multiple decision trees helps the model capture complex relationships in the data and improves the stability of the predictions.

The prediction of the Random Forest model can be represented as:

$$\mathbf{\hat{Y} = mode\ \{T_1(X), T_2(X), \ldots, T_n(X)\}}$$

where $T_1, T_2, \ldots, T_n$ represent the individual decision trees and X represents the input features. The mode denotes the class receiving the highest number of votes from the trees.

For probability-based classification, the predicted probability for a class can be expressed as:

$$\boldsymbol{P(Y = c \mid X) = \frac{1}{n} \sum_{i=1}^{n} I(T_i(X) = c)}$$

where n is the number of decision trees and I is an indicator function that takes the value 1 when the tree predicts class c, and 0 otherwise.

Random Forest was selected as one of the classification models and its performance was subsequently compared with Logistic Regression and Tuned XGBoost using different evaluation measures.

CLASSIFICATION REPORT

A classification report was generated to evaluate the performance of the classification model for the respective Diagnosis classes. It provides a detailed assessment of how effectively the model distinguishes between the different classes in the dataset. The report includes Accuracy, Precision, Recall, F1-score, and Support, allowing the performance of the model to be examined from different perspectives.

Accuracy represents the overall correctness of the model. It indicates the proportion of observations that were correctly classified out of the total number of observations. A higher accuracy value indicates that the model produces a greater number of correct predictions.

$$\textbf{Accuracy} = \frac{\textbf{TP} + \textbf{TN}}{\textbf{TP} + \textbf{TN} + \textbf{FP} + \textbf{FN}}$$

Precision:

Measures the reliability of the model's positive predictions. It represents the proportion of observations that were correctly predicted as positive among all observations predicted as positive. High precision indicates fewer false-positive predictions.

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}}$$

Recall:

Measures the ability of the model to correctly identify observations that actually belong to the positive Diagnosis class. A higher recall value indicates that the model is more effective in detecting positive cases and produces fewer false-negative predictions.

$$\textbf{Recall} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FN}}$$

F1-score:

It provides a balanced measure of Precision and Recall by combining both measures into a single value. It is particularly useful when both false-positive and false-negative predictions need to be considered while evaluating the classification model.

$$\textbf{F1-Score} = 2 \times \frac{\textbf{Precision} \times \textbf{Recall}}{\textbf{Precision} + \textbf{Recall}}$$

Support:

Represents the number of actual observations present in each Diagnosis class. It provides information about the distribution of observations across the classes and helps in interpreting the evaluation measures appropriately.

	precision	recall	f1-score	support
No PCOS	0.91	0.99	0.95	44223
PCOS	0.98	0.85	0.91	27777
accuracy			0.94	72000
macro avg	0.95	0.92	0.93	72000
weighted avg	0.94	0.94	0.93	72000

Figure 3.9: Classification Report

Figure 3.9 illustrates the classification performance of the model for the respective Diagnosis classes. The report provides Precision, Recall, F1-score, and Support for each class, enabling a detailed assessment of the model's ability to correctly classify the observations.

CONFUSION MATRIX

A confusion matrix was used to evaluate the classification performance by comparing the actual Diagnosis values with the predicted Diagnosis values. It provides a clear representation of both correctly and incorrectly classified observations and helps identify the types of errors made by the model.

The confusion matrix consists of four main components:

- True Positive (TP): Actual positive observations correctly predicted as positive.
- True Negative (TN): Actual negative observations correctly predicted as negative.
- False Positive (FP): Actual negative observations incorrectly predicted as positive.
- False Negative (FN): Actual positive observations incorrectly predicted as negative.

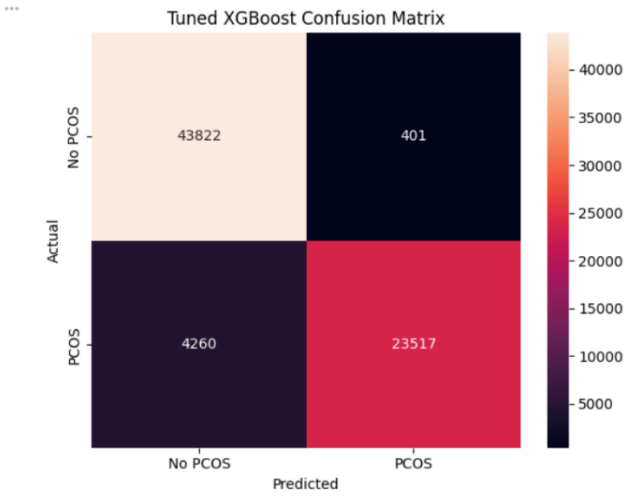


Figure 3.10: Confusion Matrix

Figure 3.10 illustrates the classification results using the confusion matrix. The matrix shows the number of correctly and incorrectly classified observations for the Diagnosis classes. It

provides an additional assessment of the model's classification behavior and helps identify the types of prediction errors produced by the model.

MODEL COMPARISON

The implemented classification models were compared to determine their **relative** predictive performance for the PCOS diagnosis prediction task. Three machine learning algorithms, namely Logistic Regression, Random Forest, and Tuned XGBoost, were considered for the comparison. Each model was trained using the prepared dataset and evaluated using the same test data to ensure a fair comparison.

The models were compared using common classification evaluation measures such as Accuracy, Precision, Recall, F1-score, and ROC-AUC. These measures provide different perspectives of model performance, including overall correctness, reliability of positive predictions, ability to identify positive cases, balance between Precision and Recall, and the model's ability to distinguish between the diagnosis classes.

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.9042	0.9027	0.8426	0.8716	0.9177
1	Random Forest	0.9353	0.9710	0.8578	0.9109	0.9321
2	Tuned XGBoost	0.9353	0.9832	0.8466	0.9098	0.9335

Figure 3.11: Machine Learning Model Comparison

Figure 3.11 illustrates the comparative performance of the implemented Machine Learning models. The comparison provides a consolidated view of the evaluation results and supports the selection of the most suitable model for the final prediction stage.

BEST MODEL SELECTION

Based on the comparative evaluation of the implemented classification models, the most suitable model was selected for generating the final prediction results. The selection was carried out by comparing the performance of Logistic Regression, Random Forest, and Tuned XGBoost using the evaluation measures obtained during model assessment.

The model demonstrating the most consistent classification performance was selected as the final prediction model. The selected model was then used to generate prediction probabilities and final class predictions for the observations. The prediction probabilities provide the likelihood of an observation belonging to the positive Diagnosis class, while the final class predictions represent the corresponding predicted Diagnosis outcome.

The generated prediction probabilities were subsequently used in the threshold optimization stage to determine an appropriate probability threshold for classification. The optimized predictions were then further utilized for risk category analysis, where the observations were grouped into different risk levels based on their predicted probability. This process helped establish a connection between the model evaluation stage and the subsequent PCOS risk prediction and dashboard analysis.

```
Best Model Based on F1 Score:
Model          Random Forest
Accuracy             0.9353
Precision             0.971
Recall               0.8578
F1 Score             0.9109
ROC-AUC              0.9321
Name: 1, dtype: object
```

Figure 3.12: Best Model Selection

Figure 3.12 illustrates the selected classification model based on the comparative evaluation results of Logistic Regression, Random Forest, and Tuned XGBoost. The model was selected by considering its performance across the evaluation metrics and its overall classification capability. The selected model was then used for the subsequent prediction and risk category analysis in the study.

4. DASHBOARD ANALYSIS

The dashboards provide a comprehensive analysis of PCOS prediction and associated risk factors by covering multiple aspects such as diagnosis, predicted PCOS likelihood, risk distribution, age, BMI, and lifestyle factors. The PCOS Prediction Overview Dashboard highlights key metrics such as total patients, PCOS predicted cases, no PCOS cases, average PCOS probability, and undiagnosed likelihood. The dashboard also presents risk distribution, actual versus predicted diagnosis, age-group distribution, PCOS probability distribution, undiagnosed PCOS likelihood, lifestyle score, and BMI-wise diagnosis. Together, these visualizations provide a clear understanding of PCOS prediction results and associated risk factors and support effective data-driven analysis.

4.1 PCOS Prediction Overview Dashboard

The PCOS Prediction Overview Dashboard provides a comprehensive overview of PCOS prediction results and important patient characteristics. It highlights key indicators such as BMI, diagnosis, and age, along with total patients, predicted PCOS cases, no PCOS cases, average PCOS probability, and undiagnosed PCOS likelihood. These indicators provide a quick summary of the prediction outcomes and overall patient distribution.

In addition, the dashboard presents the distribution of patients across different risk categories and compares actual diagnosis with predicted diagnosis. This comparison helps in understanding the performance of the prediction model and the differences between actual and predicted outcomes. Age-group analysis is used to compare patient distribution across different age categories, while the PCOS probability distribution shows how predicted likelihood varies across the patient population.

The dashboard also includes undiagnosed PCOS likelihood, lifestyle score, and BMI-wise diagnosis. These visualizations help identify patterns between patient characteristics and PCOS prediction. The combination of these charts allows different aspects of the dataset to be examined from a single dashboard, making the analysis easier and more interactive.

Overall, this dashboard provides a clear understanding of PCOS prediction outcomes and important risk-related factors. It helps in identifying variations in diagnosis and predicted likelihood across age and BMI groups. The visualizations support better interpretation of the

machine learning results and provide useful insights into the distribution of PCOS risk among patients.

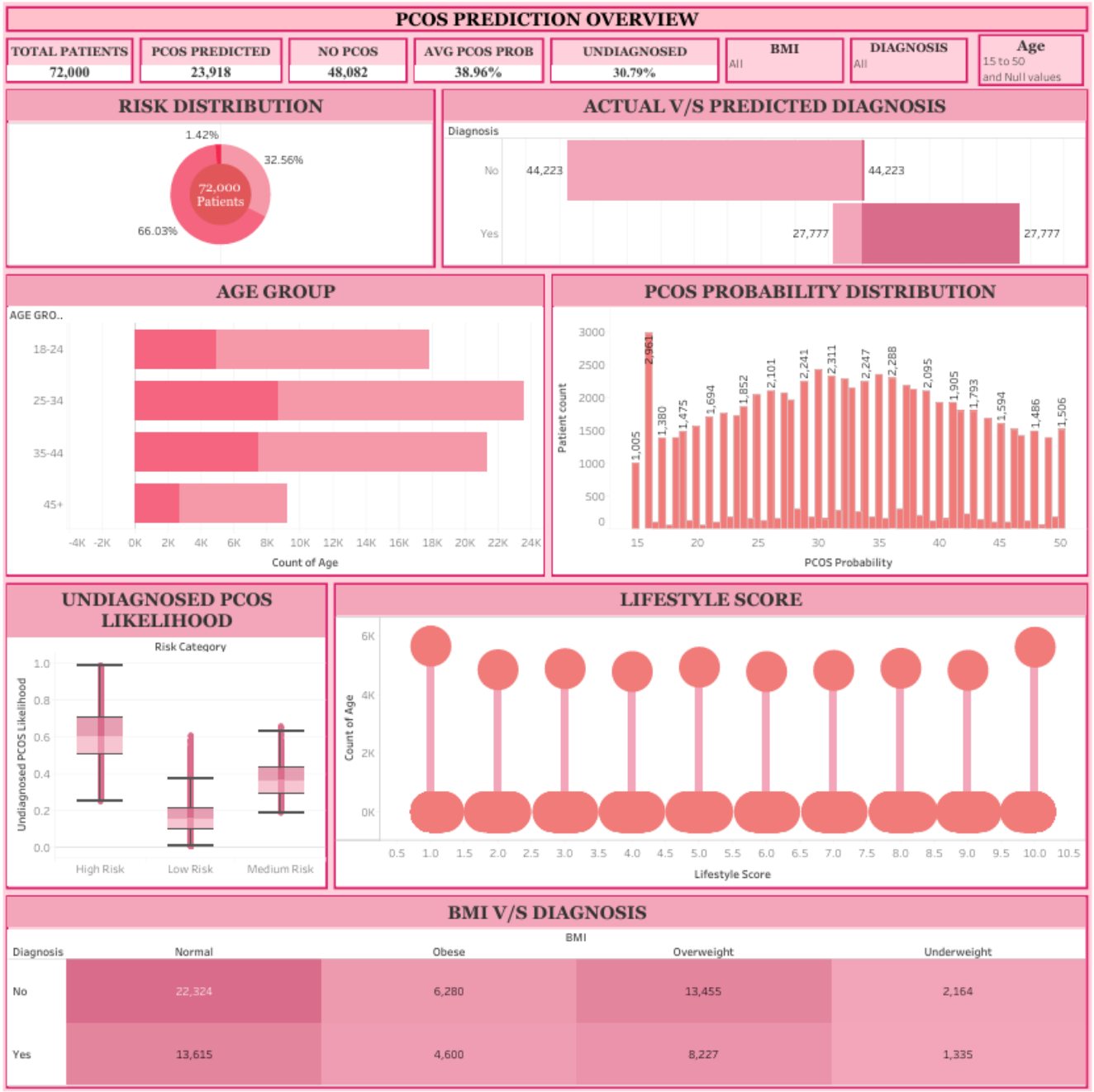


Fig 4.1.1: PCOS Prediction Overview Dashboard – Tableau

Figure 4.1.1 illustrates the PCOS Prediction Overview Dashboard, highlighting the major prediction results and patient characteristics. It displays total patients, predicted PCOS cases, no-PCOS cases, average PCOS probability, and undiagnosed likelihood. The dashboard also presents risk distribution, actual versus predicted diagnosis, age-group distribution, PCOS probability distribution, undiagnosed PCOS likelihood, lifestyle score, and BMI-wise diagnosis.

Overall Metrics (KPI Cards):

The KPI cards provide a quick summary of the PCOS prediction results. The dashboard displays the total number of patients as 72,000, with 23,918 patients predicted with PCOS and 48,082 predicted without PCOS. The average PCOS probability is 38.96%, while the undiagnosed PCOS likelihood is 30.79%. These indicators provide an overall view of the prediction results and help users quickly understand the distribution of PCOS cases. They also provide a starting point for interpreting the detailed visualizations presented in the dashboard.

BMI:

The BMI field enables analysis of patients according to different BMI categories. It helps examine the relationship between body weight classification and PCOS diagnosis. Comparing diagnosis across underweight, normal, overweight, and obese categories provides useful insight into the distribution of predicted PCOS cases among different BMI groups. This visualization can also help identify whether particular BMI categories contain a higher proportion of diagnosed or predicted PCOS cases.

Diagnosis:

The diagnosis field is used to compare patients predicted with PCOS and those predicted without PCOS. The actual versus predicted diagnosis visualization provides a comparison between the original diagnosis and model prediction, helping to understand the classification results. It also helps identify differences between actual and predicted outcomes and provides a visual representation of how the model classified the patient records.

Age:

The age field allows the patient population to be analyzed across different age groups. The dashboard categorizes patients into groups such as 18–24, 25–34, 35–44, and 45+. This helps identify the distribution of patients across age ranges and provides an opportunity to examine variations in PCOS prediction among different age groups. The visualization makes it easier to identify which age groups contain a larger number of patient records.

Risk Distribution (Donut Chart):

The risk distribution chart presents the proportion of patients classified into different PCOS risk categories. It provides a visual comparison of Low Risk, Medium Risk, and High Risk groups, making it easier to understand the overall distribution of predicted PCOS likelihood. The chart gives a quick overview of how patients are distributed across the different levels of risk.

Actual Vs Predicted Diagnosis (Butterfly chart):

The actual versus predicted diagnosis butterfly chart compares the number of patients in the No and Yes diagnosis categories. It presents the actual and predicted values on opposite sides, making it easier to identify similarities and differences between the original diagnosis and the outcomes generated by the machine learning model. This comparison helps in understanding the prediction performance and variations between actual and predicted PCOS diagnosis..

Age Group (Bar Chart):

The age-group chart displays the number of patients across different age categories. It helps identify which age groups contain a higher number of patients and provides a basis for examining age-related patterns in PCOS prediction. The comparison of age groups also supports further investigation of the distribution of PCOS cases across different stages of adulthood.

PCOS Probability Distribution (Column Chart):

The PCOS probability distribution chart shows the distribution of patients across different predicted probability levels. It helps understand how PCOS likelihood varies among the patient population and provides a broader view of the prediction probabilities generated by the model. Higher probability values indicate stronger predicted likelihood, while lower values represent comparatively lower predicted likelihood.

Undiagnosed PCOS Likelihood (Box Plot):

The undiagnosed PCOS likelihood chart presents the distribution of predicted likelihood across Low Risk, Medium Risk, and High Risk categories. It helps compare the spread and variation of predicted likelihood within each risk group. The box plot also makes it possible to observe the median, range, and variation of likelihood values for each category.

Lifestyle Score (Bubble Chart):

The lifestyle score visualization represents the distribution of patients according to their lifestyle scores. It helps examine variations in patient distribution across different lifestyle levels and supports further analysis of lifestyle-related patterns. The visualization provides an additional perspective for understanding how patient lifestyle characteristics are distributed within the dataset.

BMI vs Diagnosis (Heatmap):

The BMI versus diagnosis visualization compares PCOS diagnosis across different BMI categories. It helps identify the number of patients with and without PCOS within each BMI group and provides insight into the relationship between BMI classification and PCOS diagnosis. The heatmap makes differences between BMI categories easier to compare and supports the identification of groups with higher or lower numbers of predicted PCOS cases.

5.2 PCOS Risk Factor Analysis Dashboard

The PCOS Risk Factor Analysis Dashboard focuses on examining the major factors associated with PCOS risk among patients. It provides insights into how different demographic, clinical, and lifestyle-related characteristics influence the likelihood of developing PCOS. The dashboard includes risk categories, BMI groups, menstrual cycle patterns, insulin resistance, and other risk-related indicators to support a comprehensive analysis of factors contributing to PCOS.

The dashboard highlights key metrics such as the number of patients classified under Low Risk, Medium Risk, and High Risk categories, along with the average PCOS probability. These indicators provide a quick summary of the overall risk distribution within the dataset. Interactive filters for Age, BMI, Risk Category, and Menstrual Cycle allow users to explore risk patterns across different patient groups.

In addition, the dashboard presents visualizations that compare various risk factors and their relationship with PCOS likelihood. The Risk Factor Profile summarizes the average contribution of important clinical indicators, while additional charts analyze BMI categories, menstrual regularity, insulin resistance, and diagnosis likelihood.

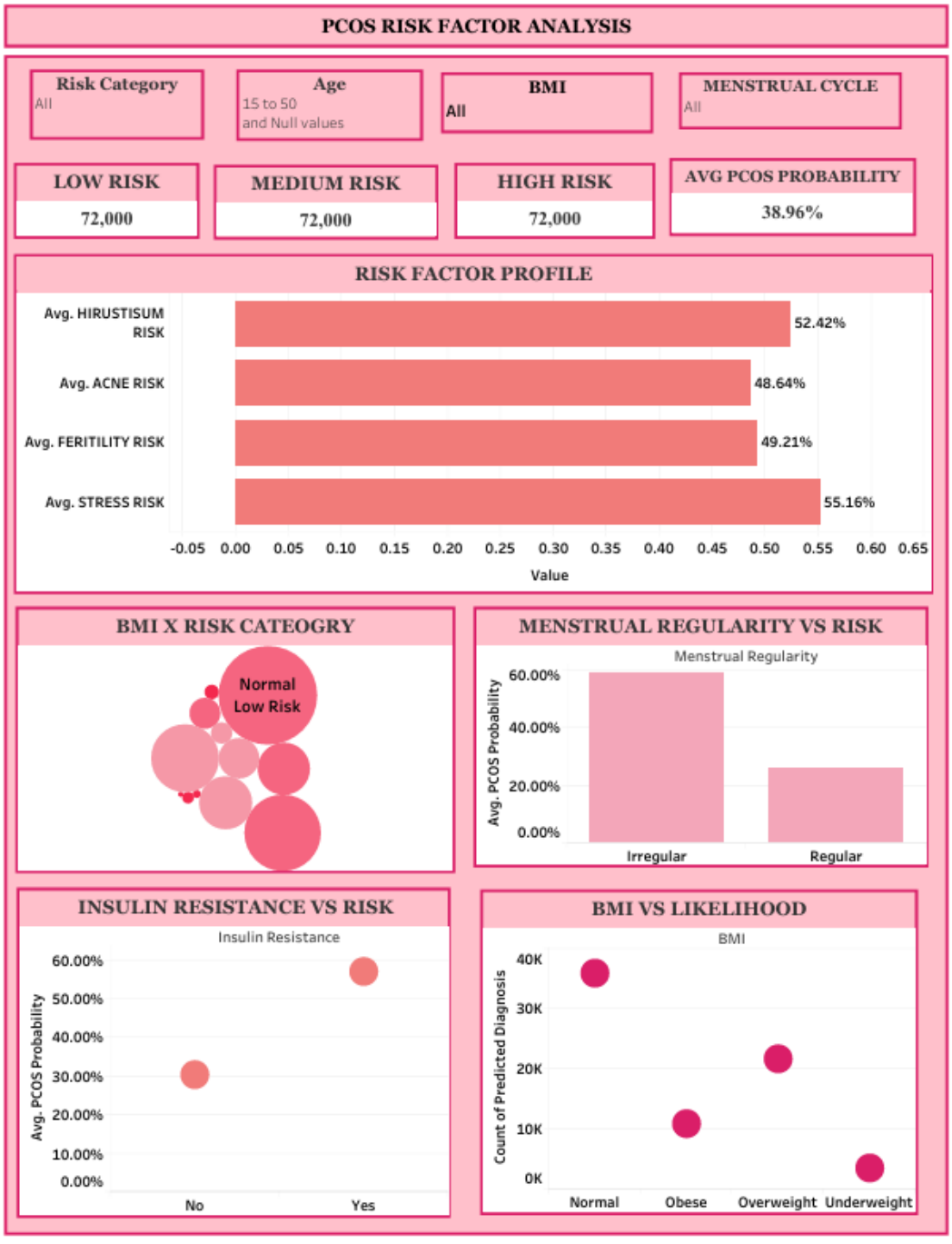


Fig 4.1.2: PCOS Risk Factor Analysis Dashboard – Tableau

Figure 4.1.2 illustrates the PCOS Risk Factor Analysis Dashboard, highlighting the major risk factors associated with PCOS likelihood. It displays risk category distribution, average PCOS probability, risk factor profiles, BMI-based risk analysis, menstrual regularity patterns, insulin resistance analysis, and BMI-wise likelihood distribution. These visualizations collectively provide insights into the factors contributing to PCOS risk and help identify important trends within the patient population.

Overall Metrics (KPI Cards):

The KPI cards provide a summary of the risk distribution within the dataset. The dashboard displays the number of patients classified under Low Risk, Medium Risk, and High Risk categories, together with the average PCOS probability. These indicators provide an overview of how patients are distributed across different levels of risk and help users quickly assess the overall risk profile of the population. The KPI cards serve as a starting point for interpreting the detailed risk-factor visualizations presented in the dashboard.

Risk Category

The Risk Category filter enables users to analyze patients according to different PCOS risk levels. Patients are categorized into Low Risk, Medium Risk, and High Risk groups based on their predicted likelihood of developing PCOS. This filter helps users focus on specific risk groups and examine how different factors vary across these categories.

Age

The Age filter allows users to examine risk patterns across different age groups. By filtering patient records based on age, it becomes possible to analyze how PCOS risk varies among younger and older individuals. This provides additional insight into age-related trends and differences in predicted PCOS likelihood.

BMI

The BMI filter supports the analysis of risk factors across different body mass index categories. It enables comparisons between Underweight, Normal, Overweight, and Obese patients to identify variations in PCOS risk. This analysis helps understand the relationship between body weight classification and the likelihood of developing PCOS.

Menstrual Cycle

The Menstrual Cycle filter allows users to explore the impact of menstrual regularity on PCOS risk. By comparing patients with regular and irregular menstrual cycles, users can identify differences in predicted PCOS probability and better understand the role of menstrual health in PCOS risk assessment.

Risk Factor Profile (Horizontal Bar Chart):

The Risk Factor Profile visualizes the average contribution of major risk-related indicators such as hirsutism risk, acne risk, fertility risk, and stress risk. The chart helps compare the relative influence of these factors on PCOS likelihood and identifies which risk indicators show higher average values within the patient population. This provides a consolidated view of important factors associated with PCOS risk.

BMI × Risk Category (Bubble Chart):

The BMI × Risk Category bubble chart illustrates the distribution of patients across different BMI and risk-level combinations. The size of each bubble represents the number of patients within a particular category, making it easier to identify BMI groups that contain higher concentrations of Low Risk, Medium Risk, or High Risk patients. This visualization helps reveal relationships between BMI classification and risk level.

Menstrual Regularity vs Risk (Column Chart):

The Menstrual Regularity vs Risk chart compares average PCOS probability between patients with regular and irregular menstrual cycles. This visualization helps determine whether menstrual irregularity is associated with higher predicted PCOS likelihood. The comparison provides valuable insight into the importance of menstrual health as a risk factor for PCOS. Patients with irregular menstrual cycles generally exhibit higher probability values, indicating a stronger association with PCOS risk. The chart also enables easy comparison of risk levels between the two groups and supports the identification of menstrual irregularity as a significant indicator in PCOS prediction.

Insulin Resistance vs Risk (Scatter Plot):

The Insulin Resistance vs Risk visualization examines the relationship between insulin resistance and predicted PCOS probability. By comparing patients with and without insulin resistance, the chart highlights differences in average likelihood values and helps identify the influence of metabolic factors on PCOS risk. This analysis supports the understanding of insulin resistance as an important contributor to PCOS. The visualization also demonstrates how metabolic conditions can affect prediction outcomes and provides a clearer view of the variation in PCOS likelihood among different patient groups. These insights help in understanding the role of insulin resistance in the development and progression of PCOS.

BMI vs Likelihood (Bubble Chart):

The BMI vs Likelihood chart compares different BMI categories based on the count of predicted diagnoses. It helps identify which BMI groups contain larger numbers of patients with predicted PCOS likelihood. This visualization provides additional insight into how body weight classification is associated with PCOS prediction outcomes and supports the identification of BMI categories that may require greater clinical attention. The size of the bubbles represents the number of patients within each BMI category, making it easier to compare their contribution to overall PCOS likelihood. The chart also helps reveal patterns between BMI levels and predicted risk, supporting a better understanding of the relationship between body weight and PCOS.

6. RESULTS AND DISCUSSION

The results of the proposed PCOS prediction framework are presented and discussed in this chapter. The performance of the machine learning models is evaluated using standard classification metrics to identify the most effective prediction model. In addition, the predicted patient outcomes are analyzed based on risk categories, demographic characteristics, BMI, and major PCOS-related risk factors. The findings provide insights into the effectiveness of the proposed framework in predicting PCOS and identifying patients with elevated risk.

Table 6.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	90.42%	90.27%	84.26%	87.16%	91.77%
Random Forest	93.53%	97.10%	85.78%	91.09%	93.21%
Tuned XGBoost	93.53%	98.32%	84.66%	90.98%	93.35%

Table 6.2 Risk Category Distribution

Risk Category	Patient Count	Percentage
Low Risk	47,539	66.03%
Medium Risk	1,019	1.42%
High Risk	23,442	32.56%
Total	72,000	100.00%

The model evaluation results presented in **Table 6.1** indicate that all three machine learning algorithms performed well in predicting PCOS. Among them, the Tuned XGBoost model achieved the highest overall performance with an accuracy of 93.53%, precision of 98.32%, and ROC-AUC score of 93.35%, making it the most suitable model for PCOS prediction. The strong classification performance demonstrates the effectiveness of machine learning techniques in identifying patients at risk of PCOS.

The prediction analysis of 72,000 patient records identified 23,918 patients with predicted PCOS and 48,082 patients without PCOS. The average PCOS probability was 38.96%, while the

undiagnosed PCOS likelihood was 30.79%. Risk categorization results presented in **Table 6.2** show that 66.03% of patients belong to the Low-Risk group, whereas 32.56% fall under the High-Risk category. This indicates that a substantial portion of the population exhibits characteristics associated with elevated PCOS risk.

Patient demographic analysis revealed that the 25–34 age group represents the largest share of the dataset. The age distribution suggests that PCOS-related characteristics are more frequently observed among women in their reproductive years. The diagnosis comparison further showed consistency between predicted outcomes and diagnosis categories, indicating reliable model performance.The probability distribution analysis demonstrated variations in predicted PCOS likelihood across patients. While a large proportion of individuals were associated with moderate probability values, a considerable number of patients exhibited higher likelihood scores, supporting the effectiveness of the risk classification framework.

BMI-related analysis highlighted noticeable differences across weight categories. The Normal BMI group contained the highest number of patients, followed by Overweight and Obese groups. The BMI versus diagnosis and BMI versus likelihood analyses suggest that body weight plays an important role in PCOS prediction and risk assessment.The risk factor evaluation identified Stress Risk (55.16%) as the most influential factor, followed by Hirsutism Risk (52.42%), Fertility Risk (49.21%), and Acne Risk (48.64%). These findings indicate that both hormonal symptoms and lifestyle-related conditions contribute significantly to PCOS risk.

Clinical factor analysis further revealed that patients with irregular menstrual cycles exhibited considerably higher PCOS probability compared to those with regular cycles. Similarly, individuals with insulin resistance showed increased likelihood of PCOS when compared with non-insulin-resistant patients.

Overall, the results demonstrate that the proposed PCOS prediction framework successfully combines machine learning and visual analytics to identify potential PCOS cases and evaluate associated risk factors. The findings emphasize the importance of BMI, menstrual regularity, insulin resistance, stress, fertility-related issues, acne, and hirsutism in understanding PCOS risk, while the Tuned XGBoost model provides accurate and reliable prediction performance.

7. CONCLUSION

The PCOS Prediction and Risk Factor Analysis study provides an effective approach for identifying potential PCOS cases and analyzing the factors associated with the condition. By combining machine learning techniques with interactive visual analytics, the study successfully evaluates patient characteristics, prediction outcomes, and risk patterns within the dataset. The findings demonstrate the value of data-driven methods in supporting PCOS analysis and risk assessment.

The machine learning evaluation showed that all three classification models achieved strong predictive performance. Among them, the Tuned XGBoost model produced the best results with an accuracy of 93.53%, precision of 98.32%, and ROC-AUC score of 93.35%. These results confirm the effectiveness of machine learning algorithms in accurately predicting PCOS and identifying individuals at higher risk.

The dashboard analysis provided valuable insights into patient distribution, diagnosis patterns, age groups, BMI categories, and PCOS probability levels. Risk categorization revealed that most patients belonged to the Low-Risk group, while a significant proportion were classified as High Risk. The visualizations helped present complex prediction results in a clear and interactive manner.

The risk factor analysis identified stress, hirsutism, fertility-related issues, acne, menstrual irregularity, insulin resistance, and BMI as important contributors to PCOS risk. Patients with irregular menstrual cycles and insulin resistance showed higher PCOS probability, highlighting the influence of hormonal and metabolic factors on the condition.

Overall, the study demonstrates that the proposed framework effectively combines machine learning and visual analytics to predict PCOS and evaluate associated risk factors. Future work may focus on incorporating larger datasets, additional clinical variables, and advanced predictive techniques to further improve prediction accuracy and risk assessment.

REFERENCES

Azziz, R. et al. (2016). Polycystic Ovary Syndrome. Nature Reviews Disease Primers, 2(1), 16057.

Escobar-Morreale, H. F. (2018). Polycystic Ovary Syndrome: Definition, Aetiology, Diagnosis and Treatment. Nature Reviews Endocrinology, 14(5), 270–284.

Teede, H. J. et al. (2018). Recommendations from the International Evidence-Based Guideline for the Assessment and Management of Polycystic Ovary Syndrome. Human Reproduction, 33(9), 1602–1618.

Kotu, V., & Deshpande, B. (2019). Data Science: Concepts and Practice (2nd ed.). Morgan Kaufmann.

Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.

Hastie, T. et al. (2017). The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd ed.). Springer.

Pedregosa, F. et al. (2011). Scikit-Learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794.

Chawla, N. V. et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321–357.

Tableau Software LLC. (2024). Tableau Desktop User Guide. Tableau Documentation.

James, G. et al. (2021). An Introduction to Statistical Learning with Applications in R (2nd ed.). Springer.

Han, J., Kamber, M., & Pei, J. (2022). Data Mining: Concepts and Techniques (4th ed.). Morgan Kaufmann.

APPENDIX

```
# Install required libraries

!pip install -q xgboost imbalanced-learn

# Import required packages

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split, RandomizedSearchCV

from sklearn.metrics import (

    accuracy_score,

    precision_score,

    recall_score,

    f1_score,

    roc_auc_score,

    confusion_matrix,

    classification_report)

from sklearn.linear_model import LogisticRegression

from sklearn.ensemble import RandomForestClassifier

from xgboost import XGBClassifier

from imblearn.over_sampling import SMOTE

# LOAD DATASET
```

```
# Read PCOS dataset

df = pd.read_csv("pcos_prediction_dataset.csv")

# Display dataset information

print(df.shape)

print(df.head())

# DATA PREPROCESSING

# Separate features and target variable

X = df.drop("Diagnosis", axis=1)

y = df["Diagnosis"]

# Convert diagnosis labels into numerical values

y = y.map({"No": 0, "Yes": 1})

# Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X,y,

    test_size=0.30,random_state=42,stratify=y)

# Handle missing values in numerical columns

numeric_cols = X_train.select_dtypes(include=np.number).columns

for col in numeric_cols:

    median_value = X_train[col].median()

    X_train[col] = X_train[col].fillna(median_value)

    X_test[col] = X_test[col].fillna(median_value)

# Handle missing values in categorical columns

categorical_cols = X_train.select_dtypes(exclude=np.number).columns

for col in categorical_cols:
```

```
    mode_value = X_train[col].mode()[0]
    X_train[col] = X_train[col].fillna(mode_value)
    X_test[col] = X_test[col].fillna(mode_value)
# Convert categorical variables into numerical format
X_train_encoded = pd.get_dummies(X_train)
X_test_encoded = pd.get_dummies(X_test)
# Align train and test columns
X_train_encoded, X_test_encoded = X_train_encoded.align(
    X_test_encoded,
    join="left",
    axis=1,
    fill_value=0)
# HANDLE CLASS IMBALANCE USING SMOTE
smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(
    X_train_encoded,
    y_train)



# LOGISTIC REGRESSION MODEL
lr = LogisticRegression(
    max_iter=1000,
    random_state=42)
```

```
lr.fit(X_train_smote, y_train_smote)
lr_pred = lr.predict(X_test_encoded)
lr_prob = lr.predict_proba(X_test_encoded)[:, 1]
# RANDOM FOREST MODEL
rf = RandomForestClassifier(
    n_estimators=300,
    random_state=4)
rf.fit(X_train_smote, y_train_smote)
rf_pred = rf.predict(X_test_encoded)
rf_prob = rf.predict_proba(X_test_encoded)[:, 1]
# XGBOOST MODEL WITH HYPERPARAMETER TUNING
xgb = XGBClassifier(
    objective="binary:logistic",
    eval_metric="logloss",
    random_state=42)
param_grid = {
    "n_estimators": [100, 200, 300],
    "max_depth": [3, 5, 7],
    "learning_rate": [0.01, 0.05, 0.1]}
random_search = RandomizedSearchCV(
    estimator=xgb,
    param_distributions=param_grid,
    n_iter=15,
```

```
    scoring="f1",
    cv=3,
    random_state=42,
    n_jobs=-1)
random_search.fit(
    X_train_smote,
    y_train_smote)
best_xgb = random_search.best_estimator_
# Generate predictions
xgb_pred = best_xgb.predict(X_test_encoded)
xgb_prob = best_xgb.predict_proba(X_test_encoded)[:, 1]
# MODEL EVALUATION
# Calculate evaluation metrics
accuracy = accuracy_score(y_test, xgb_pred)
precision = precision_score(y_test, xgb_pred)
recall = recall_score(y_test, xgb_pred)
f1 = f1_score(y_test, xgb_pred)
roc_auc = roc_auc_score(y_test, xgb_prob)

# Display model performance
print("Accuracy :", accuracy)
print("Precision :", precision)
print("Recall :", recall)
```

```
print("F1 Score :", f1)

print("ROC-AUC :", roc_auc)

# CONFUSION MATRIX

cm = confusion_matrix(y_test, xgb_pred)

print("Confusion Matrix")

print(cm)

# RISK CATEGORY CREATION

# Create risk categories based on prediction probability

risk_category = pd.cut(

    xgb_prob,

    bins=[0.00, 0.30, 0.70, 1.00],

    labels=["Low Risk", "Medium Risk", "High Risk"],

    include_lowest=True)

# EXPORT DATA FOR TABLEAU DASHBOARD

Tableau_data = df.loc[X_test.index].copy()

Tableau_data["Actual_Diagnosis"] = y_test.values

Tableau_data["Predicted_Diagnosis"] = xgb_pred

Tableau_data["PCOS_Probability"] = xgb_prob

Tableau_data["Risk_Category"] = risk_category

# Export dataset for Tableau visualization

Tableau_data.to_csv(

    "PCOS_Tableau_Dataset.csv",

    index=False)
```